

Ragavendra Raja J

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Computer Science Engineering Core

Vellore, India

2021- present

- Cumulative GPA : **8.77/10.0**

Relevant Coursework: Data Structures and Algorithms, OOP, Operating Systems, Database Systems, Computer Networks, Data Science, Web Mining, Machine Learning, Software Engineering, Advanced Python Programming, Cloud Computing, Cryptography & Networks Security

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, HTML/CSS, JavaScript

Frameworks: Next.js, TailwindCSS, Framer Motion, Flask, Apache Maven, Node.js, MongoDB

Developer Tools: Git, Github, VSCode, Linux, Docker, Jenkins

Libraries: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, NLTK, Tkinter

PROJECTS

DataSecurity in Lightweight Cryptography for Secure and Decentralized Data Storage

Ongoing

- Designed an advanced cryptographic model integrating **ChaCha20** encryption with **Ikeda chaotic maps** to enhance data security in IoT environments.
- Implemented tile-based image encryption, achieving **99.6%** NPCR and passing NIST randomness tests for robustness.
- Integrated IPFS-based blockchain for decentralized data storage, ensuring immutability, tamper resistance, and content-based addressing.

Advanced Hotel Recommendation System with Web Scraping and Sentiment Analysis

- Engineered an AI-powered hotel recommendation system by leveraging advanced web scraping techniques to extract and process data from online sources.
- Applied sentiment analysis to over 5,000 user reviews from **Booking.com**, utilizing NLP techniques to generate data-driven insights and enhance recommendation accuracy.
- Designed a robust data pipeline utilizing **BeautifulSoup** and **NLTK** for sentiment-driven ranking of customer feedback.

Phishing Site Detection Using Machine Learning

- Evaluated multiple machine learning algorithms, including **Multilayer Perceptron (MLP)**, **Random Forest**, **Decision Tree**, **SVM**, to enhance phishing site classification accuracy
- Extracted and analyzed critical URL-based features such as domain attributes, redirection behavior, and protocol usage to improve detection efficiency. Conducted a comprehensive performance analysis using accuracy, precision, recall, and **AUC-ROC metric** to identify the most effective phishing detection model.

EXTRACURRICULAR AND EXPERIENCE

Student Manager | *graVITas'24 (Technical Fest) & Riviera'25 (Cultural Fest)*

Sep. 2024 – Feb. 2025

- Managed sales operations for Riviera'24 & Riviera'25, one of India's top 5 cultural fests, overseeing sponsorship acquisition, event ticket sales, and merchandise distribution.

Senior Core Committee Member | *DigitSquad VIT*

Jan. 2022 – Jan. 2024

- Planned, organized, and executed 5+ technical events and workshops, increasing member engagement and enhancing hands-on technical skill development.

Core Member | *Youth Red Cross, VIT*

Mar. 2022 – Feb 2024

- Coordinated 2 successful donation drives, mobilizing over 300 community members to contribute essential goods and services, resulting in a significant boost in local engagement and measurable social impact.

CERTIFICATIONS

Microsoft Certified: **Azure AI Fundamentals**

Software Engineering Job Simulation Goldman Sachs

Game Theory Stanford University

SQL Injection Attacks EC Council

Problem Solving (Intermediate) HackerRank